

Exclusions:

Chronic lung disease
Pneumonia, croup, bronchiolitis
Cardiac disease
Airway anomalies
Neurologic disease
Anaphylaxis
FB aspiration

UK Pediatric ED Asthma Pathway

For children ≥ 2 years of age with acute asthma exacerbation



1. Vitals obtained in triage
2. Examine patient (H&P)
3. PRAM score patient.
(add up scores for each criteria)

PRAM score (MDCalc) Pediatric Respiratory Assessment Measure		
Criteria	Description	Score
Oxygen saturation (on room air)	$\geq 95\%$	0
	92-94%	1
	$< 92\%$	2
Suprasternal retractions	Absent	0
	Present	2
Scalene muscle contractions	Absent	0
	Present	2
Air entry by auscultation (If asymmetry, use the most severely affected lung field)	Normal	0
	Decreased at bases	1
	Widespread (apex and base)	2
	Minimal or absent	3
Wheezing ≥ 2 auscultation zones must be affected (If asymmetry, use the most severely affected lung field)	Absent	0
	Expiratory only	1
	Inspiratory $\pm$ expiratory	2
	Audible without stethoscope Or silent chest	3

PRAM 0-3
Mild

PRAM 4-7
Moderate

PRAM 8-9
Severe

PRAM 10-12
Respiratory Failure

Treatment algorithm based on PRAM score

PRAM 0-3
Mild

Albuterol 4-8 puffs MDI

Dexamethasone 0.6 mg/kg oral solution (max 16 mg)

PRAM 4-7
Moderate

Ipratropium-albuterol Duo-Neb 0.5-2.5 mg/3 mL (continuous = 3 Duo-Nebs, order 9 mL)

Dexamethasone 0.6 mg/kg oral solution (max 16 mg)

Consider budesonide 1 mg

PRAM 8-9
Severe

Ipratropium-albuterol Duo-Neb 0.5-2.5 mg/3 mL (continuous = 3 Duo-Nebs, order 9 mL)

Magnesium sulfate 50 mg/kg IV (max 2g) with 20 mL/kg IV fluid bolus

Methylprednisolone 1-2 mg/kg IV (max dose 60mg)

Consider budesonide 1 mg

PRAM 10-12
Respiratory Failure

Notify ED attending immediately

Ipratropium-albuterol Duo-Neb 0.5-2.5 mg/3 mL (continuous = 3 Duo-Nebs, order 9 mL)

Magnesium sulfate 50-75 mg/kg IV (max 2g) with 20 mL/kg IV fluid bolus

Methylprednisolone 1-2 mg/kg IV (max dose 60mg)

Consider IM epinephrine 0.01 mg/kg (max dose 0.5mg)

Consider budesonide 1 mg



Repeat PRAM score 1 hour after intervention

Repeat PRAM score immediately after intervention

PRAM 0-3
Mild

Plan for **discharge** if PRAM has remained ≤ 3

Discharge Criteria

- On room air
- PRAM score 0-3
- Asthma/MDI Education completed
- Follow-up in 1-2 days
- PCP, Asthma clinic, ED discharge clinic
- Consider pediatric pulmonology referral
- Prescriptions
- HMPC complete

PRAM 4-5
Moderate

Monitor another hour and repeat PRAM score

- If worse, move to next appropriate path
- If PRAM still 4-5, begin albuterol 8 puffs MDI Q2 hours and **admit to pediatrics floor**
 - RT score Q2 hours
- Plan for **discharge** if PRAM score ≤ 3 for an hour

PRAM 6-7
Moderate-Severe

Continuous albuterol 20mg/hr

Magnesium sulfate 50 mg/kg IV (max 2g) with 20 mL/kg IV fluid bolus

Repeat PRAM 1 hour after intervention

- If better or worse, move to next appropriate path
- If PRAM still 6-7, begin continuous albuterol and **admit to pediatrics PCU**
 - RT score Q1 hour

PRAM 8-9
Severe

Continuous albuterol 20mg/hr

Give IV magnesium if haven't already.

Consider IM epinephrine 0.01 mg/kg (max dose 0.5mg).

Repeat PRAM after intervention

- If better or worse, move to next appropriate path
- If PRAM remains ≥ 8 , begin continuous albuterol and **admit to PICU**
 - RT score Q1 hour

PRAM 10-12
Respiratory Failure

Continuous albuterol 20mg/hr

Give 2nd dose of IV magnesium.

Consider IM epinephrine if hadn't already

Begin NIPV

Admit to PICU
RT score Q1 hour

NIPV
Titrate to work of breathing and hypoxemia
Initial settings

- IPAP 10/EPAP 5
- IPAP 14/EPAP 6
- IPAP 16/EPAP 8
- Max settings: IPAP 20/EPAP 10

Inpatient Asthma Protocol

Acute Care

Revised: 01/04/2024

Assess patient upon admission and complete PRAM Score (treatment initiation based on score)
Complete Asthma Control Test (ACT)
Resume home controller medications; assess compliance with home medications and need to escalate
Start discharge planning and teaching

PEDIATRIC RESPIRATORY ASSESSMENT MEASURE (PRAM) SCORING		
Criteria	Description	Score
Oxygen Saturation	≥ 95%	0
	92 to 94%	1
	< 92%	2
Suprasternal Retractions	Absent	0
	Present	2
Scalene Retractions	Absent	0
	Present	2
Air Entry	Normal	0
	Decreased at Bases	1
	Decreased Widespread (Apex and Base)	2
	Minimal or Absent	3
Wheezing	Absent	0
	Expiratory Only	1
	Inspiratory ± Expiratory	2
	Audible without Stethoscope or Silent Chest	3

Inclusion: pediatric patients ≥ 2 years of age with acute asthma exacerbation

Exclusion: patients with following diagnosis or past medical history: chronic lung disease; pneumonia, croup, bronchiolitis; cardiac disease; airway anomalies; neurologic disease; anaphylaxis; foreign body aspiration

Available through MDCalc: <https://www.mdcalc.com/calc/3384/pediatric-respiratory-assessment-measure-pram-asthma-exacerbation-severity>

PRAM 0-3 Mild Location: Floor

Albuterol 8 puffs q4 hours

PRAM Score by RT q4 hours

If PRAM 0-3 x 2 consecutive scores notify first call provider to change to 4 puffs q4 hours and prepare for discharge

Steroids: continue Prednisolone 1 to 2 mg/kg divided twice daily (max 60 mg) x 3-5 days if started in ED
Discuss if re-dosing Dexamethasone 0.6 mg/kg (max 16 mg)

Discharge Planning Checklist:

- ☐ On Room Air
- ☐ PRAM Score 0-3
- ☐ Asthma Education Completed
- ☐ Asthma Action Plan Completed
- ☐ PCP Follow Up Scheduled 1 to 2 Days Post-Discharge and/or Pulmonary Asthma Clinic
- ☐ Discharge Prescriptions Delivered from Meds2Beds

PRAM 4-5 Moderate Location: Floor

Albuterol 8 puffs q2 hours

PRAM Score by RT q2 hours

Steroids: continue Prednisolone 1 to 2 mg/kg divided twice daily (max 60 mg) x 3-5 days if started in ED vs re-dose Dexamethasone 0.6 mg/kg (max 16 mg)

Consider Pediatric Pulmonology consult to discuss need for inpatient consult or outpatient follow up in Asthma Clinic

Reassessment
at 2 hours

Treatment based off PRAM score:

Mild (PRAM 0-3): notify first call provider to space to albuterol 8 puffs q4 hours and follow mild pathway

Moderate (PRAM 4-5): continue albuterol 8 puffs q2 hours

Moderate - Severe (PRAM 6-7): notify first call provider to escalate to continuous albuterol and follow moderate-severe pathway

PRAM 6-7 Moderate to Severe Location: Progressive Care Unit

Continuous albuterol 20mg/hr x 1 hour
Consider ipratropium 500 mcg q4 hour x 24 hours

PRAM Score by RT q1 hour

Order telemetry and make patient NPO

Steroids: continue Prednisolone 1 to 2 mg/kg divided twice daily (max 60 mg) x 3-5 days if started in ED vs re-dose Dexamethasone 0.6 mg/kg (max 16 mg)

Consider Pediatric Pulmonology consult to discuss need for inpatient consult or outpatient follow up in Asthma Clinic

Reassessment
at 1 hour

PRAM ≥ 8

PRAM 6-7

Transfer to PICU
Follow Severe/Failure
Pathway
(Consider calling rapid
response, notify first call
provider)

Begin 2nd continuous albuterol 20mg/hr

Moderate-Severe (PRAM 6-7) after 4 hours continuous albuterol, transfer patient to PICU

Note:

- 4 to 6 puffs of albuterol with spacer = 2.5 mg albuterol by nebulization (one (1) nebulization solution vials)
- 8 puffs of albuterol with spacer = 5 mg albuterol by nebulization (two (2) nebulization solution vials)

Treatment Based on PRAM Score
Notify Provider to Escalate OR De-Escalate Treatment per PRAM Score
Start Supplemental Oxygen for Oxygen Saturations ≤ 90%, Titrate per Protocol

Inpatient Asthma Protocol PICU

Revised: 01/04/2024

PRAM 8-12 Severe /Failure Location: PICU

Start/Change to methylprednisolone 1mg/kg (max 60 mg) q6 hours - first dose STAT if not given in previous treatment location (ER, Floor, Progressive Care Unit)

Continuous albuterol 20mg/hr + ipratropium 500 mcg q4 hours x 24 hours
Magnesium sulfate bolus 50 mg/kg (max 2000 mg) with 20 mL/kg fluid bolus

PRAM Score by RT on PICU arrival and q1 hour

Order telemetry and make patient NPO

Consider Pediatric Pulmonology consult

Reassessment
at 1 hour

PRAM 8-9 or lower

Continue current treatment. De-escalate per acute care pathways as able and transfer to floor when PRAM Score is 5

PRAM 10-12

Continuous albuterol 20mg/hr

Consider IM epinephrine 0.01 mg/kg (max 0.5 mg) or re-dosing Magnesium bolus

Begin NIPV- titrate to work of breathing and degree of hypoxemia

Initial settings: IPAP 10/EPAP 5
IPAP 14/EPAP 6
IPAP 16/EPAP 8
Max settings: IPAP 20/EPAP 10

PEDIATRIC RESPIRATORY ASSESSMENT MEASURE (PRAM) SCORING		
Criteria	Description	Score
Oxygen Saturation	≥ 95%	0
	92 to 94%	1
	< 92%	2
Suprasternal Retractions	Absent	0
	Present	2
Scalene Retractions	Absent	0
	Present	2
Air Entry	Normal	0
	Decreased at Bases	1
	Decreased Widespread (Apex and Base)	2
	Minimal or Absent	3
Wheezing	Absent	0
	Expiratory Only	1
	Inspiratory ± Expiratory	2
	Audible without Stethoscope or Silent Chest	3

Inclusion: pediatric patients ≥ 2 years of age with acute asthma exacerbation

Exclusion: patients with following diagnosis or past medical history: chronic lung disease; pneumonia, croup, bronchiolitis; cardiac disease; airway anomalies; neurologic disease; anaphylaxis; foreign body aspiration

Available through MDCalc: <https://www.mdcalc.com/calc/3384/pediatric-respiratory-assessment-measure-pram-asthma-exacerbation-severity>

Other Treatments to consider if no improvement or worsening clinical status:

Magnesium:
Infusion: 10 mg/kg/hr, titrate to goal magnesium level 3-5

Terbutaline:
Bolus: 2 to 10 mcg/kg (max 400 mcg)
Infusion: 0.1 to 0.4 mcg/kg/min titrated by clinical response up to 3 mcg/kg/min

Ketamine:
Bolus: 0.5 to 2 mg/kg
Infusion: 1 to 2 mg/kg/hr

Consider ECMO consult if intubating

Treatment Based on PRAM Score
Notify Provider to Escalate OR De-Escalate Treatment per PRAM Score
Start Supplemental Oxygen for Oxygen Saturations ≤ 90%, Titrate per Protocol